FRESHSENSE AI

A Multimodal Edge-AI Framework for Food Freshness Assessment and Shelf-Life Prediction Using Vision, VOC Sensing and OCR

PROJECT REPORT

Student	Dhriti Mishra
Register number	24BDE0102
Project type	Individual academic prototype
Report format	Black and white | 23 October 2026

Prototype status: working local demonstration; OCR, vision, sensor, and safety claims are limited to the evidence stated in this report.

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The final section is References, with the two base papers clearly identified.

Aim of the Project

The aim of FreshSense AI is to develop a practical web application that helps users capture food-package information, read date labels with OCR where available, review suggested product details, and track each purchased batch in a household pantry. The project explores how computer vision, package text, optional environmental measurements, and user corrections can be organized into an explainable freshness record. Its recommendations are informational and are not certified food-safety judgments.

- Read package text and common expiry phrases such as EXP, best before, use by, and sell by.
- Suggest a food category and make recognition evidence visible for user review.
- Ask the user to confirm or enter a missing date and record purchased quantity.
- Save separate purchase batches and track their date, storage context, and reminders.
- Keep the project honest about incomplete models, illustrative sensor values, and validation limits.

Project Title

Formal title: A Multimodal Edge-AI Framework for Food Freshness Assessment and Shelf-Life Prediction Using Vision, VOC Sensing and OCR

Product name: FreshSense AI

Title-scope note: shelf-life regression and physical ESP32 calibration are future-phase work. The current prototype supports a vision prototype, OCR/date parsing, a deterministic decision workflow, and a sensor integration abstraction; it does not provide a validated shelf-life prediction model.

Abstract

FreshSense AI is a responsive web prototype for scanning packaged foods and maintaining a batch-specific pantry inventory. A user can submit a package image, receive candidate text/date/category information, review or correct it, enter the purchased quantity, and save the purchase for later tracking. The system consists of a React progressive web app, a FastAPI backend, SQLAlchemy data models, a local SQLite demo option, OCR/date parsing, an interpretable image-analysis prototype, and a rule-based assessment flow. A Longitudinal Freshness Passport links repeat scans and records evidence, missing inputs, conflicts, and user-confirmed details.

The prototype demonstrates an integrated user workflow rather than a validated food-safety instrument. OCR depends on a locally available engine; category matching and image freshness classification have limited evidence; sensor values can be illustrative unless supplied by calibrated hardware. The strongest defensible contribution is the combination of reviewable scan evidence with a separate pantry batch ledger. In a small image experiment documented by the repository, an inactive candidate reached 52.6% validation accuracy and 53.1% macro recall on a fixed sample split; the score is exploratory and is not used to claim real-world performance. Package label accuracy and learned multimodal fusion have not been benchmarked.

Keywords: food freshness, OCR, computer vision, VOC sensing, food inventory, human-in-the-loop AI, Freshness Passport

Register Number

No.	Student name	Register number
1	Dhriti Mishra	24BDE0102

The project information available for this report identifies one team member. Add any additional members here if the submission team has changed.

Students Name

Dhriti Mishra (Register Number: **24BDE0102**)

Team size recorded for this report: one student. Names and registration numbers are reproduced from the details provided for the report.

Tools Used

Area	Tools / technologies	Use in the project
Frontend	React 18, TypeScript, Vite, Tailwind CSS, React Router	Responsive scan, result, login, dashboard, history, and pantry screens.
Frontend data	TanStack Query, Axios, Zod	API requests, cached server state, and response/input validation.
Progressive web app	vite-plugin-pwa, service worker, Web App Manifest	Installable app shell and camera/upload workflow.
Backend	Python, FastAPI, Pydantic, Uvicorn	Versioned REST API and request/response schemas.
Database	SQLAlchemy, Alembic, SQLite local demo, PostgreSQL configuration	Persistence and schema migration support.
AI / image	Pillow, interpretable image features, logistic-regression candidate	Prototype image analysis; trained candidate remains inactive.
Text recognition	Tesseract OCR (optional system installation), Python OCR/date parser	Extract image text when available and parse labeled dates.
Sensors	Temperature/RH/VOC API abstraction; ESP32-ready interface	Accept environmental context; physical sensor and calibration are future work.
Development	Git, npm, Python virtual environment, Vitest	Source control, package management, and frontend verification.

Highlights OR Few Features of Your Project

- **Freshness Passport:** an evidence-aware scan record that identifies which inputs contributed and what is missing or uncertain.
- **Longitudinal item history:** repeat scans can be linked to a named food item and displayed chronologically.
- **Expiry label understanding:** OCR text is searched for multiple date labels; ambiguous or missing dates remain for user review.
- **Human confirmation:** users can confirm or correct category/date values before tracking a purchase.
- **Batch-level pantry:** each purchase has its own quantity, date provenance, storage location, opened date, notes, and reminder window.
- **Explainable use-next cue:** pantry records are ordered by entered/confirmed package dates with the basis explained; this is a rule, not a learned predictor.
- **Privacy-aware image retrieval:** uploaded photos are served through an authenticated, user-scoped, non-cacheable endpoint.
- **Phone-ready PWA:** camera capture and an installable shell support local phone demonstrations on the same network.

Novelty statement: the applied contribution is the connection of scan provenance and user corrections to longitudinal item history and separate purchase-batch tracking. This is a product/workflow contribution and a research hypothesis, not evidence of scientific priority, food-safety accuracy, or waste reduction.

Project Modules

Module	Main responsibility
1. User and access	Sign-in, registration, user-scoped scans, and local demo accounts.
2. Capture and upload	Phone camera/image upload, framing guidance, and retake path.
3. Vision assessment	Prototype image features or disclosed fallback; no calibrated probability or safety claim.
4. OCR and expiry parser	Optional Tesseract text extraction and labeled-date parsing.
5. Product/category recognition	Candidate category and product-name identification from available package text.
6. Sensor/context input	Temperature, humidity, and VOC data handling through a provider abstraction.
7. Decision and assessment	Deterministic, missing-modality-safe rules combine available evidence and surface limitations.
8. Freshness Passport	Evidence, confidence/calibration caveats, missing inputs, conflicts, model identity, and repeat-scan timeline.
9. Pantry batch ledger	Purchase quantity, date source, storage, opened date, notes, and reminder settings.
10. Date watch and history	Inventory sorting, scan history, in-app reminders, and opt-in browser alerts while the app is open.

Module Description

Capture and upload

The user takes a picture or uploads a package image. The interface supports mobile capture and displays scan evidence on a result screen. Clear product and date framing, focus, and lighting improve the chance of OCR success.

OCR and expiry extraction

When Tesseract is installed and available, image text is extracted. The parser prioritizes recognized label cues such as EXP, expiry, best before, use by, consume by, sell by, and valid until. It avoids treating a labeled manufacture/production date as an expiry date. If no reliable date is available, the application asks the user to confirm, enter, or explicitly leave the date unknown.

Category and vision

The package category is suggested from recognized product text and configured mappings. A separate freshness classifier uses interpretable image features when a reviewed local artifact exists; otherwise the app discloses a prototype fallback. The candidate model has not been activated because its limited validation results do not justify promotion.

Assessment and Passport

The service layer records which modalities contributed to a scan and computes deterministic assessment output. The Passport keeps scan observations and limitations together. Temperature/humidity can be contextual only, and VOC rules use an explicit fixed threshold until calibrated hardware is available.

Pantry and reminders

A saved purchase is a distinct batch. The user can manage quantity, package date and its source, category, storage location, opened date, notes, and reminder timing. In-app date watch remains visible; browser alerts are checked while FreshSense is open and do not run after the app is closed.

Role of Team Members

Team member	Role / responsibility
Dhriti Mishra (24BDE0102)	Individual project lead: requirements, application design, frontend/backend implementation, OCR and AI workflow integration, data/inventory flow, verification, documentation, and demonstration.

This role summary reflects a single-member project as provided. If any part of the implementation or documentation was completed by a teammate or supervisor, revise the allocation before formal submission.

Software Description

Overall architecture

FreshSense AI follows a client/API/data-layer design. The React/Vite PWA communicates with FastAPI routes under /api/v1. Routes delegate to services for OCR, vision, sensors, expiry interpretation, decisions, and assessments. SQLAlchemy models persist users, tracked items, scans, OCR results, sensor readings, predictions, assessments, model versions, and pantry inventory. The local demo can use SQLite; PostgreSQL is supported through configuration and Alembic migrations.

Layer	Software description
Client	React + TypeScript single-page application with mobile camera/upload and PWA shell.
API	FastAPI JSON endpoints for authentication, scans, vision/OCR, sensor context, tracked items, and inventory.
Services	Python service layer keeps parsing, deterministic assessment, and data access outside frontend screens.
Persistence	SQLAlchemy ORM models; SQLite is used by the local self-contained demo; PostgreSQL schema is managed with Alembic.
AI integrations	Optional OCR executable; image-feature prototype/model interface; explicit unimplemented status for learned fusion and shelf-life regression.

Local demo accounts

The repository seeds demo@example.com and reviewer@example.com on first SQLite demo startup. Both local-only accounts use password FreshSenseDemo2026!. These are demonstration credentials and must not be reused or published on a real deployment.

Development status

Recorded checks include frontend TypeScript/build verification and four UI tests, backend syntax compilation, and an inventory API smoke check (create returned HTTP 201 and a subsequent update succeeded). Backend integration tests with live migrations and physical phone/browser tests were not recorded as completed. The package-label benchmark has no collected validation manifest, so OCR/category recognition accuracy is unknown.

Coding

The project uses TypeScript/React for the client and Python/FastAPI for the API and AI-related services. Code is organized into pages/components, API hooks, routes, repositories, OCR/ML/sensor providers, and services. The following simplified excerpt is adapted from the implemented expiry parser. It accepts a date only when it matches a recognized expiry label, and it guards against interpreting manufacturing dates as expiry dates.

```
        for match in _EXPIRY_LABEL.finditer(text or ""):
            expiry = _parse(match.group(1))
            if expiry:
                return {
                    "expiry_date": expiry,
                    "date_detected": True,
                    "expiry_label": label,
                    "expiry_text": match.group(1).strip(),
                }
```

```
    if _MANUFACTURE_LABEL.search(text or ""):
        return {"expiry_date": None, "date_detected": False, ...}
```

The implementation handles labeled and several numeric/textual date formats. A date-only fallback is intentionally restricted when multiple dates are visible, and partial day/month stamps are retained as evidence without inventing a year. OCR errors still occur; users must review candidate results and provide corrections. The full parser is in `backend/app/ocr/date_parser.py`; the OCR provider is in `backend/app/ocr/service.py`.

Project verification summary

Check	Recorded evidence	Status
Frontend build and UI	TypeScript check/build passed; four UI tests passed in the recorded implementation pass.	Passed for recorded checks
Backend syntax	Python syntax compilation passed.	Passed for recorded check
Inventory API	Local smoke check created a purchase and updated its fields successfully.	Path smoke-checked
OCR/category quality	No labeled package-photo validation benchmark collected.	Not measured
Classifier	52.6% accuracy, 53.1% macro recall on a fixed 445-image validation split; candidate inactive.	Exploratory only
Hardware/mobile	No physical sensor calibration or comprehensive physical-phone smoke test recorded.	Not validated

Screenshot of the Errors Segments

The screenshot supplied for this project shows an OCR/category failure. The package photo visibly contains a best-before stamp, but the result screen displayed “Product not identified” and “Printed expiry date: Not detected.” This documents a recognition miss on that image; it is not a controlled benchmark result.

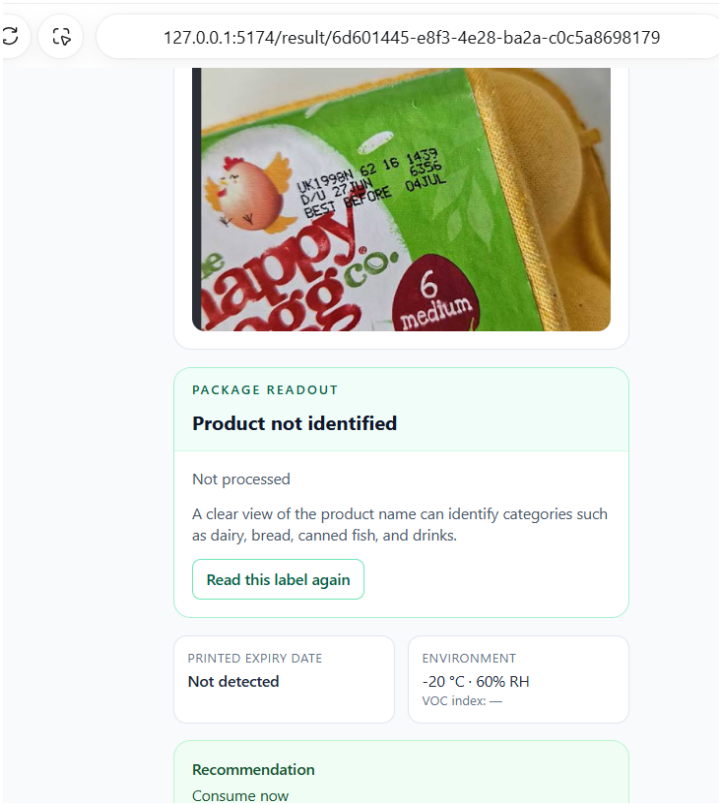


Figure 1. Actual FreshSense result screenshot supplied by the student; package stamp visible, product/date not recognized.

Screenshot of the Error Recovered Segments

This live FreshSense scan-result screenshot shows the package photo and OCR evidence for its printed BEST BEFORE stamp. The parser preserved “04JUL” as a partial date and disclosed that the year was not read. The Passport keeps the partial evidence visible without inventing a full expiry date; the user can confirm or enter the date before saving a batch to the pantry.

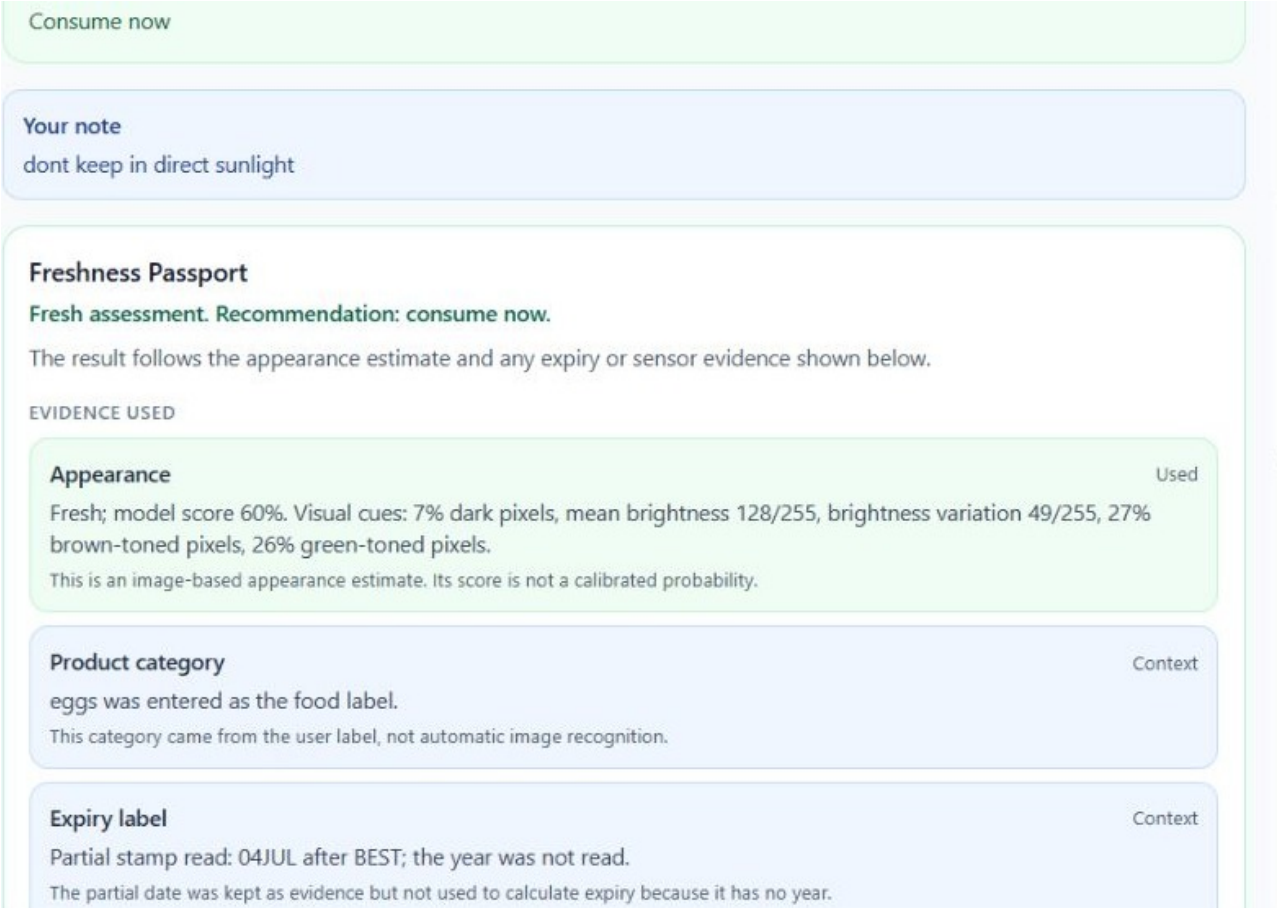


Figure 2. Live in-app-browser capture: partial BEST BEFORE date retained as evidence; the year remains unconfirmed.

What was recovered: The date parser now recognizes common expiry phrases and preserves a partial stamp for review. When the year or a complete date is missing, FreshSense avoids calculating an expiry date and leaves the user a path to confirm or enter it. OCR quality still requires a labeled evaluation set.

Possible Future Work

1. Create a consented, representative package-label dataset covering date formats, categories, brands, lighting, glare, and phone cameras; report OCR/date and category metrics on held-out package groups.

2. Add crop/rotate, blur/glare detection, multi-frame capture, and language-aware OCR; surface raw text and calibrated confidence before confirmation.

3. Expand category recognition with a curated taxonomy and evaluate per-class precision, recall, F1, and a confusion matrix.

4. Complete the Freshness Passport event log with image hash/reference, raw OCR, field-level source, confidence, user edits, model version, and batch timeline.

5. Calibrate ESP32-compatible temperature, humidity, and VOC sensors; store device identity, units, calibration date, and sampling interval.

6. Collect repeated, paired image/date/sensor observations and compare image-only output with rule-based Passport fusion on item-grouped held-out data.

7. Train and independently evaluate a lightweight visual model and a shelf-life regressor before enabling them in the application.

8. Add reliable background reminders, household sharing controls, backup/restore, audit logging, and privacy retention/deletion controls.

9. Test on real phones and browsers, including offline/network recovery, camera permissions, accessibility, and realistic upload sizes.

10. Run a usability study comparing manual entry, isolated scans, and the Passport workflow using task time, record completeness, corrections, and user feedback.

Conclusion

FreshSense AI is an integrated student prototype that connects package capture, OCR/date parsing, category suggestions, user confirmation, Freshness Passport evidence, and batch-based pantry tracking. Its strongest presentable feature is the human-reviewed longitudinal record: a scan can remain useful even when an automated field is missing because the user can confirm or enter the value and continue managing the purchase batch.

The project has a defensible applied workflow contribution, but not yet proven scientific novelty, 100% accuracy, food-safety prediction, shelf-life regression, or sensor fusion performance. The failure screenshot documents a real recognition miss, while the live result capture shows a partial expiry stamp retained for user confirmation. Priorities are a labeled package benchmark, representative phone testing, calibrated hardware, and independent evaluation.

References - Base Papers and Base Links

The two papers most directly connected to this prototype are marked **BASE PAPER**. Their methods provide background for OCR and sensor-based freshness assessment; they do not validate the FreshSense implementation.

[BASE PAPER 1 - OCR] Smith, R. (2007). An Overview of the Tesseract OCR Engine. In *Proceedings of the Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)*, Vol. 2, pp. 629-633. IEEE. DOI: https://doi.org/10.1109/ICDAR.2007.4376991. Base link: IEEE Xplore record.

[BASE PAPER 2 - VOC / SENSOR FRESHNESS] Wang, M., Gao, F., Wu, Q., Zhang, J., Xue, Y., Wan, H., & Wang, P. (2018). Real-time assessment of food freshness in refrigerators based on a miniaturized electronic nose. *Analytical Methods, 10*, 4741-4749. DOI: https://doi.org/10.1039/C8AY01242C. Base link: Royal Society of Chemistry article page.

[RELATED MULTIMODAL STUDY] Senge, J. M., Kaltenecker, F., & Krupitzer, C. (2025). Integrating Sensor Data, Laboratory Analysis, and Computer Vision in Machine Learning-Driven E-Nose Systems for Predicting Tomato Shelf Life. *Chemosensors, 13*(7), 255. DOI: https://doi.org/10.3390/chemosensors13070255. Base link: Publisher article page.

[VISION DATASET] Risdin, F., Salauddin, H., Tufa, T. A., & Hossain, M. J. (2026). *AgriFreshNET Freshness and Shelf-Life Image Dataset* (Version 1) [Data set]. Mendeley Data. DOI: https://doi.org/10.17632/42m5tb7yv9.1. Dataset link: Mendeley Data record.

[VISION DATASET] Abayomi-Alli, O., & Damaševi■ius, R. (2022). *Fruit Quality Datasets (FruQ-DB)* [Data set]. Zenodo. DOI: https://doi.org/10.5281/zenodo.7224690. Dataset link: Zenodo record.

The two base papers are the OCR engine overview and the electronic-nose freshness study. The 2025 multimodal study is additional context. Dataset records are cited because they support the exploratory image-model experiment. This is a focused project bibliography, not a systematic literature review.